CSE 404: Intro to Machine Learning
Team Project

CSE 404 will end with a semester-long team project paper and presentation (instead of a final exam). Groups should consist of about 5 students. The project consists of four main components: proposal, short paper, long paper, and group presentation.

Project Proposal (Topic, Dataset, & Team Info)
Due: February 10

Groups will propose the research task they aim to solve, which can be any ethical ML task. The written proposal must include:

- An outline of what your group is planning to do for this project. Include descriptions of the dataset(s) you are planning to use, as well as the model(s) you tentatively plan to use and how you will evaluate your results. Please see the last page for more details.
- Motivate why this is an interesting ML problem worth exploring. Will it contribute a better model? A better modeling pipeline? Help interpret a socially important phenomenon? Predict some interesting real-world events?
- Include a literature review to set the project in context. What has been done before? How will your project be different? How will it improve on the current work?
- Bios of the researchers! Include your name, NetID, department, and year in the program. Also list what each team member will likely be responsible for or how you plan to divide the work fairly.
- This initial proposal should be roughly similar to the *Abstract, Introduction, Related Works*, and *Dataset/Methodology* sections of a typical conference paper submission.
- Length: 2 pages, ACL format

Short Paper (Models)
Due: March 12

Groups will submit their implemented models, initial experimental results, and describe their planned experimental analysis. This report should add any new related works, modify the abstract and introduction as needed, and expand on the methodology section by adding more details of the models and initial experimental results.

Length: 4 pages; similar to a conference short paper.

LONG PAPER (FINAL REPORT) & PRESENTATION SLIDES
DUE: APRIL 9

Groups will submit their completed project. The final version of your paper should include a description of everything done for this project: literature comparison, data collection, annotation, processing, model and/or pipeline development and technical details, thorough experimental analysis, future work, and conclusion.

Length: 8 pages; similar to a conference long paper.

GROUP PRESENTATIONS
APRIL 14–23

Each group will present their findings through a PowerPoint presentation. The length of time for each presentation will be announced after I have a count of the number of groups. Currently I have reserved 4 class days for presentations, but this may be modified if the number of groups changes. We may need to use the final exam day for presentations, depending on the number of groups.

ACCOUNTABILITY

In addition to the Short Paper and Final Paper, you will be required to submit *evaluation forms* (will be available on D2L & Piazza). These forms can include as much or as little detail as you feel is needed. Use this form to communicate any problems that the team is facing, e.g., a team member who is not contributing to the project in a timely manner, or at all. You can also use the forms to let me know if someone is going "above and beyond" for the project.

SUGGESTIONS FOR IDEAS

Datasets: Papers with Code

Check out past work at some top conferences:

ACL IJCAI EMNLP AAAI ICWSM NeurIPS KDD COLING

SemEval releases trending NLP tasks each year, usually with datasets.

Dataset Warnings

When choosing your dataset, please keep in mind that data collection can become the most time-consuming part of your project. It would be fastest to find a pre-existing dataset that you can use directly or with simple modifications. However, you must double check that the dataset is safe for use. If you plan to collect and annotate (label) your own data, this could end up taking too long depending on your task.

Model Requirements

Everyone will be required to implement baseline models and a more complex neural network model (i.e., not a Feedforward NN). For the baseline models, you may choose from any of the models discussed in class, but the most relevant model for the task based on your literature review is preferred. For the neural network model, you should choose whichever architecture is best suited for your task. For example, tasks using language-based features might choose RNNs or Transformer-based NNs, while tasks using images might choose Convolutional Neural Networks (CNNs). For evaluation, results should be reported for all models using at least the four basic metrics: precision, recall, accuracy, and F_1 score. All code for the project must be submitted through D2L or GitHub.

Project Proposals That Are Usually Rejected

In general, I accept most project ideas. However, project proposals that fall under the following categories may be rejected.

- Harmful and/or unethical ideas
- Projects completed or in progress for other classes, capstones, work, or research
- Projects from prior semesters "borrowed" from friends
- Kaggle (or similar) datasets and tasks
- Pre-existing works with only minor changes